

SIMATS ENGINEERING

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404 & DSA0405

CO 3 - ASSESSMENT TOOL 2: Data Interpretation

Name : Y. Jayesh Ranjan

Reg No : 192472201

COURSE OUTCOME COVERED

- Apply data preparation and exploratory data analysis techniques, including data summarization, data distribution analysis, outlier treatment, and measures of asymmetry, to preprocess and interpret datasets effectively.
- Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.

Sl.No 1-10

1. Data Summarization

A dataset of 11 employee salaries (in ₹'000) is: 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120.

(a) Compute the mean and median. (b) Explain why the mean is a poor measure of central tendency here. (c) Which measure of dispersion would you trust more — standard deviation or IQR — and why?

2. Outlier Treatment

Using the dataset above, apply the $1.5 \times \text{IQR}$ rule to identify outliers. Show Q1, Q3, IQR, and the lower/upper fences. Which values qualify as outliers? If you cap them at the fences (Winsorization) instead of removing them, how would the mean change qualitatively?

3. Measuring Asymmetry

Two datasets have the same mean (50) and standard deviation (10), but Dataset A has Pearson's skewness coefficient = +1.2 and Dataset B has -0.8. Sketch (describe) the shape of each distribution and state which is more likely to have salary-like data with a long right tail.

4. Coefficient of Skewness Calculation

For a dataset: Mean = 62, Median = 58, Mode = 52, SD = 12. Calculate skewness using (a) Karl Pearson's coefficient (mode method), (b) Karl Pearson's coefficient (median method, $Sk = 3(\text{Mean} - \text{Median})/\text{SD}$). Compare the two results and comment on why they might differ slightly.

5. Variance vs Standard Deviation Interpretation

Two production lines report these stats:

Line A: Mean = 200 units/day, Variance = 64

Line B: Mean = 150 units/day, Variance = 36

Which line has greater relative variability? Use the Coefficient of Variation to justify your answer, not just raw variance.

Sl.No 11-20

6. Sampling Distribution Reasoning

A population has mean $\mu = 500$ and SD $\sigma = 80$. A random sample of $n = 64$ is drawn.

(a) Compute the standard error of the sample mean. (b) What is the probability that the sample mean exceeds 520, assuming the sampling distribution is approximately normal? (c) If n were increased to 256, how does the standard error change, and what principle explains this?

7. Covariance Calculation and Sign Interpretation

Given paired data on advertising spend (X, ₹'000) and sales (Y, ₹ lakh):

X: 10, 20, 30, 40, 50

Y: 15, 18, 25, 30, 42

Calculate $\text{Cov}(X, Y)$. Is the relationship direction consistent with intuition? What does the magnitude alone fail to tell you (that correlation would)?

8. Correlation vs Causation Trap

A city reports that ice cream sales and drowning incidents have a correlation coefficient of $r = 0.85$. A newspaper claims "ice cream causes drowning." Identify the statistical flaw, name the likely confounding variable, and explain what additional evidence (not just correlation) would be needed to claim causation.

9. Effect of Outliers on Correlation

A dataset of 9 points shows $r = 0.42$ (weak positive). One extreme outlier is removed, and r becomes 0.89. What does this reveal about the fragility of Pearson's correlation? Would Spearman's rank correlation be more robust here? Justify.

10. Data Preparation — Missing Values

A dataset of 200 rows has 15% missing values in the "Income" column, and analysis shows these missing values are more common among unemployed respondents (not random). (a) Classify this missing data mechanism (MCAR/MAR/MNAR). (b) Why would mean imputation introduce bias here? (c) Suggest a better strategy.

Sl.No 21-30

11. Standardization/Normalization

A feature "Age" ranges from 18–75 and another feature "Income" ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm. Explain, with reasoning (not just definitions), why failing to standardize these variables would distort analysis, and calculate the z-score for Age = 45 if mean = 40 and SD = 12.

12. Distribution Shape from Summary Stats Alone

A dataset has Mean = 45, Median = 45, Mode = 45, and skewness ≈ 0 , but kurtosis = 5.8 (leptokurtic). Explain what this combination tells you about the distribution's shape that skewness and central tendency alone would miss.

13. Grouped Data — Mean and Variance

Class Interval: 0–10, 10–20, 20–30, 30–40, 40–50

Frequency: 5, 8, 15, 12, 10

Calculate the mean and variance using the step-deviation method. Show the assumed midpoints and working.

14. Comparing Two Distributions via CV

Portfolio A: Mean return = 12%, SD = 3%

Portfolio B: Mean return = 18%, SD = 5%

An investor claims Portfolio B is "riskier in absolute terms but not necessarily worse per unit of return." Evaluate this claim quantitatively using Coefficient of Variation and explain what CV captures that SD alone doesn't.

15. Outlier's Effect on Variance

A dataset of 10 values has variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Without recalculating exactly, explain directionally and reason about the magnitude of distortion to (a) the mean, (b) the variance — and why variance is affected disproportionately more than the mean.

Sl.No 31-40

16. Continuous Distribution — Area Interpretation

A continuous random variable X is normally distributed with $\mu = 100$, $\sigma = 15$.

(a) What proportion of values lie between 85 and 130? (b) A value of 145 is observed — how many standard deviations away is it, and would you flag it as a potential outlier under the empirical rule?

17. Sampling Bias vs Sampling Error

A survey estimating average household expenditure samples only online panel respondents.

The sample mean is ₹18,000 vs a known population mean of ₹14,500 (from census data). (a)

Is this discrepancy better explained by sampling error or sampling bias? (b) Would increasing the sample size fix this problem? Justify.

18. Covariance Sign vs Correlation Strength

Two variable pairs both show positive covariance: Pair 1 has Cov = 850, Pair 2 has Cov = 12.

Explain why you cannot conclude Pair 1 has a "stronger relationship" than Pair 2 without additional information, and state what that missing information is.

19. Estimation — Confidence Interval Reasoning

A sample of $n = 100$ items has mean = 250 and SD = 40. Construct a 95% confidence interval for the population mean. Then explain, in interpretation (not formula) terms, what this interval does and does NOT tell you about the true population mean.

20. Integrated Case Study

A retail company's daily sales data (in ₹'000) for 15 days is:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

(a) Identify the outlier and justify using both the IQR method and z-score method.

(b) Compute mean and median with and without the outlier — compare the sensitivity of each.

(c) Compute skewness with the outlier included — classify the distribution shape.

(d) Recommend whether the company should remove, cap, or investigate the outlier before reporting monthly performance, and justify your recommendation using the statistical evidence from (a)–(c).

11. Standardization / Normalization

Given

Age:

Income:

We need to explain why standardization is required and calculate the z-score for:

Given:

The question specifically asks for the reasoning and the z-score.

CO3-AT2

Formula for Z-score

Where:

- = observed value
- = mean
- = standard deviation
- = standardized value

The z-score tells us **how many standard deviations a value is away from the mean.**

Calculation

Given:

Substitute:

Therefore,

Interpretation

Age 45 is approximately **0.42 standard deviations above the mean age of 40.**

Why standardization is needed here

Age ranges only from **18 to 75**, whereas Income ranges from **₹20,000 to ₹20,00,000.**

For a distance-based algorithm, the numerical size of the variables affects the calculated distance.

Income has much larger numerical values, so without standardization, **Income would dominate the distance calculation**, while Age would have comparatively less influence.

After standardization, both variables are put on a comparable scale.

Final Answer

Standardization is required because Income has a much larger numerical scale than Age and would otherwise dominate a distance-based analysis.

12. Distribution Shape from Summary Statistics Alone

Given

The question asks what these values tell us about the distribution.

CO3-AT2

Step 1: Check central tendency

We have:

When all three are equal, the distribution is approximately **symmetric around 45**.

Step 2: Check skewness

Given:

A skewness close to zero means there is **little or no asymmetry**.

So the distribution is approximately symmetric.

Step 3: Check kurtosis

Given:

The question states that this is **leptokurtic**.

A leptokurtic distribution has relatively **heavy tails**, meaning there is a greater possibility of extreme observations compared with a normal distribution.

Important interpretation

The mean, median and mode being equal tells us about the **centre and symmetry**.

But they do not tell us much about the **tails**.

That is where kurtosis is useful.

Final Answer

The distribution is approximately **symmetric and centered around 45** because:

and

However,

indicates a **leptokurtic distribution with relatively heavy tails**. Therefore, the distribution can be symmetric while still having a greater tendency for extreme values.

13. Grouped Data — Mean and Variance

This is the main numerical problem in your set. The PDF specifically requires the **step-deviation method**, including assumed midpoints and working.

CO3-AT2

Given Data

Class Interval	Frequency
0–10	5
10–20	8
20–30	15
30–40	12
40–50	10

Step 1: Find the midpoint

Formula:

Where:

- = lower class limit
- = upper class limit
- = midpoint

For 0–10:

For 10–20:

Similarly:

So:

Class		
0–10	5	5
10–20	8	15
20–30	15	25
30–40	12	35
40–50	10	45

Step 2: Find total frequency

Step 3: Select assumed mean

Take the middle midpoint:

Class width:

Now use:

Here, is the step-deviation value.

Step 4: Prepare the table

Class					
0–10	5	5			20
10–20	8	15			8
20–30	15	25	0	0	0
30–40	12	35	1	12	12

Class					
40–50	10	45	2	20	40
Total	50			14	80

Therefore:

Step 5: Calculate Mean

Formula

Where:

- = assumed mean
- = class width
- = sum of frequency $\times$ step deviation
- = total frequency

Substitute:

Therefore:

Mean = 27.8

Step 6: Calculate Variance

Formula

Where:

- = variance
- = class width
- = sum of
- = total frequency

Substitute:

Final Answer

14. Comparing Two Distributions via CV

This is another direct numerical question. The PDF gives the means and SDs and asks you to evaluate the investor's statement using **Coefficient of Variation**.

CO3-AT2

Given

Portfolio A

Portfolio B

Formula

Where:

- **CV** = coefficient of variation
- **SD** = standard deviation
- **Mean** = average return

CV tells us the **amount of risk relative to the average return**.

Step 1: CV of Portfolio A

Step 2: CV of Portfolio B

Step 3: Compare

Portfolio	Mean	SD	CV
A	12%	3%	25%
B	18%	5%	27.78%

Portfolio B has:

So B has greater **absolute variability/risk**.

But:

So B also has greater **relative variability**.

Conclusion

The investor says Portfolio B is riskier in absolute terms but not necessarily worse per unit of return.

From the CV calculation:

Therefore, **Portfolio A actually has lower relative risk.**

So based purely on CV, Portfolio A provides the better risk-to-return variability.

Final Answer

Portfolio B is riskier in absolute terms **and also has slightly higher relative risk** according to CV.

15. Outlier's Effect on Variance

The question asks for a **directional explanation without exact recalculation**, so we should not unnecessarily perform a complete variance calculation.

CO3-AT2

Given

Number of observations:

Original variance:

Original value:

Mean is approximately:

Incorrect value:

(a) Effect on Mean

The value changes from:

Increase in the value:

Since there are 10 observations, the increase in the mean is approximately:

Original mean:

Approximate new mean:

So:

The exact new mean cannot be determined from the information given because the exact original dataset is not provided.

(b) Effect on Variance

Variance is calculated using squared deviations.

Formula

Where:

- = individual observation
- = mean
- = number of observations
- = squared deviation from the mean

Originally, the value 40 is close to the mean 42:

Squared deviation:

So its contribution to variance is small.

But after the recording error:

Its squared deviation is approximately:

This is extremely large compared with:

Therefore, the variance will increase **very dramatically**.

Why is variance affected more than the mean?

The mean uses the original value directly:

So the error of 360 is simply distributed among 10 observations.

But variance uses:

The deviation is **squared**.

Therefore, a very large error becomes much larger when squared.

Simple comparison

Original deviation:

Squared:

Incorrect value's approximate deviation:

Squared:

This clearly shows why variance is much more sensitive to the outlier.

Final Answer

- **Mean:** increases substantially.
- **Variance:** increases enormously.
- **Reason:** variance uses squared deviations, so the extreme value 400 has a disproportionately large effect.